

Rohit Prajapat

Bachelor of Technology

Mathematics and Computing

Sardar Vallabhbhai National Institute of Technology, Surat

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EDUCATION

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat

2024 – 2028

B.Tech in Mathematics & Computing CGPA: 8.40/10 (Till 4th Sem)

SKILLS

Data Analysis and Visualization : NumPy, Pandas, Matplotlib, Power BI

Math Foundations: Linear Algebra, Probability & Statistics, Calculus, Hypothesis Testing

Programming Languages: Python, C, C++, JavaScript (ES6), SQL

EXPERIENCE

Indian Institute of Technology (IIT) Mandi

May 2026 – July 2026

Summer Research Intern

Mandi, Himachal Pradesh

- Developing a Clinical Ultrasound Report Generation system using Deep Learning under the guidance of Dr. Deepak Raina.
- Designing and training a CNN-based ResNet model to automatically analyze ultrasound images and generate structured diagnostic reports.
- Working on medical image preprocessing pipelines and integrating AI-driven inference to assist radiologists in clinical workflows.

Indore Smart City Development Limited (ISCDL)

June 2025 – July 2025

Data Analytics Intern

Indore, India

- Developed interactive dashboards using Power BI, Excel, and Python to visualize city-level and traffic data.
- Analyzed traffic data to identify congestion hotspots and automated data processing workflows.
- [GitHub](#) [Report](#)

PROJECTS

Bhu-Nirakshak – AI Land Encroachment Detection | [Live](#) | [GitHub](#)

Smart City Hackathon

- *Tech Stack: Python, OpenCV, GIS, Satellite/Drone Imagery*
- Built an AI system for illegal construction detection using computer vision and building footprint extraction models.
- Leveraged satellite/drone imagery to identify and flag unauthorized structures in urban zones.

RAG-Based Knowledge Assistant | [GitHub](#)

- *Tech Stack: Python, FastAPI, LangChain, ChromaDB, Ollama (LLaMA3), React, Vite, TailwindCSS, SentenceTransformers*
- Built a fully local RAG (Retrieval-Augmented Generation) assistant supporting PDF, PPTX, DOCX, CSV and TXT uploads with semantic search over documents — no API key required.
- Implemented a FastAPI backend with LangChain orchestration, ChromaDB vector store, and local LLM inference via Ollama, with streaming responses and source attribution.
- Developed a responsive React + Vite frontend with real-time chat interface and document ingestion pipeline using SentenceTransformers embeddings.

Medical Ultrasound Image Caption Generation

IIT Mandi Summer Internship

- *Tech Stack: Python, PyTorch, SigLIP, Transformer Decoder*
- Designed and implemented a vision-language model to generate captions from ultrasound images, combining a SigLIP vision encoder with a Transformer-based text decoder.
- Processed and trained on a dataset of 3,261 medical images using transfer learning and image augmentation techniques.
- Achieved improved caption quality through beam search decoding, evaluated using BLEU-4, ROUGE-L, METEOR, and CIDEr metrics.